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Chancellor for Research and Graduate Education at the University of Wisconsin –Madison, and the Wisconsin Alumni Research Foundation (SKC).

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ABSTRACT

Purpose: This study investigated the effects of a 6-week eccentrically-biased training intervention between Romanian deadlift (RDL) and Nordic hamstring exercises (NHE) on regional hamstring muscle architecture and morphology. **Methods:** Participants were randomly allocated to an RDL or NHE intervention group and trained twice per week for 6 weeks followed by a 2-week detraining period. Biceps femoris long head (BFlh) architecture (fascicle length [FL] and pennation angle [PA]) and muscle thickness (MT) were measured using extended-field-of-view ultrasound. Anatomical cross-sectional areas (ACSA) of BFlh, semimembranosus, and semitendinosus were also assessed via ultrasound. Muscle architecture and morphology were measured proximally, in the middle, and distally along the thigh. Separate linear mixed effects models were run for each outcome. **Results:** Thirty-two healthy participants (15 females, 17 males) aged 20.8 (0.9) years were enrolled and allocated to RDL (n = 16) or NHE (n = 16) groups. Following the intervention, BFlh FL increased (0.80 [0.6, 1.0] cm (9%), $p < 0.001$), while PA decreased uniformly across all regions (-1.2° [-1.5, -0.9] (10%), $p < 0.001$) and MT did not change ($p = 0.29$). Both BFlh FL and PA returned to baseline after detraining. No differences were observed between groups for changes in FL ($p = 0.55$), PA ($p = 0.74$), or MT ($p = 0.48$). Hamstrings ACSA increased after the intervention (0.78 cm^2 [0.45, 1.11] (10%), $p < 0.001$) and remained elevated after detraining. There were no significant differences in ACSA between groups after the intervention ($p = 0.60$). **Conclusions:** Changes in FL, PA, and ACSA are consistent between an eccentrically-biased RDL and NHE intervention. These findings provide practitioners choice in exercise selection for eliciting adaptations relevant to hamstring injury prevention. **Key Words:** BICEPS FEMORIS, CROSS-SECTIONAL AREA, FASCICLE LENGTH, PENNATION ANGLE, RESISTANCE TRAINING, ULTRASOUND

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INTRODUCTION

Hamstring strain injuries are some of the most prevalent injuries in running-based sports, including soccer, American football, rugby, and track and field (1–4). The burden of hamstring strain injuries (HSI) is also high in these sports, considering the time loss from training and competition, cost associated with injured athlete salary and care, and implications for team success (5–7). Many preventative measures have been implemented in an attempt to minimize the prevalence and burden of HSI (8–12). Eccentric hamstring strength training is one of the most widely investigated preventative interventions, which can induce physiological adaptations thought to be protective of strain-induced damage experienced during the injurious terminal swing phase of sprinting (13, 14).

The Nordic hamstring exercise (NHE) is an eccentric hamstring strength training intervention proven to reduce hamstring injury rates in sport (15). This evidence-based hamstring injury prevention intervention also elicits protective adaptations in the muscle architecture of the commonly injured biceps femoris long head (BFlh), including increased fascicle length (FL) and decreased pennation angle (PA) (15–18). These adaptations in BFlh muscle architecture are desirable as FL and PA are both modifiable risk factors for hamstring injury (19, 20). It is theorized that the increase in FL accompanied by a concurrent decrease in PA is due to the addition of sarcomeres in series which subsequently reduce the propensity for sarcomere overstretching in activities such as sprinting (21, 22).

Despite the effectiveness of the NHE in modifying BFlh muscle architecture and reducing hamstring strain injury risk (19, 23, 24), this evidence-based intervention is often not implemented

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in practice (8). Athletes have negative attitudes towards the NHE due to the high level of delayed onset muscle soreness and stiffness after performing the NHE (8, 25). The efficacy of the neutral hip position when performing the NHE has also been questioned by many researchers and practitioners, as this loads the bi-articular hamstring, including the BFlh at relatively short muscle lengths (26–29). In contrast, hip dominant hamstring exercises such as the 45-degree hip extension and Romanian deadlift (RDL) may allow for hamstring loading at longer muscle lengths, which may also preferentially load the often-injured BFlh (17, 30–33).

Although hip-dominant exercises can elicit BFlh architectural changes, most studies involving the RDL do not eccentrically bias the hamstring like the NHE (34–36). Recent evidence suggests that different contraction durations (i.e., time under tension) may influence regional hypertrophy across the quadriceps muscles (37). Additionally, preventative architectural and morphological changes (e.g., anatomical cross-sectional area, ACSA) are also observed with as little as four weeks of eccentric training (16, 38). These findings suggest that not only matching load, but also contraction-type times, may be important to consider when comparing different exercises and that adaptations can be achieved after short training durations.

Previous research has also focused on measuring hamstring muscle architecture at the mid-belly only (16, 38). This approach fails to account for the known variation in architecture along the length of the hamstring muscles (39–42) or the non-uniform lengthening, activation, and adaptations induced by resistance training (43–46). Due to the high rate of injury within the proximal muscle-tendon junction of the BFlh and non-uniform muscular adaptation to resistance training, the RDL seems particularly suited to target this commonly injured muscle, which has

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relatively high surface electromyography activity during this exercise, particularly in the proximal region of this muscle (35, 47).

Despite the evidence supporting the use of eccentric hamstring strength training to elicit beneficial architectural and morphological adaptations, it is unknown if architectural adaptations induced by hip-dominant versus knee-dominant exercises are more pronounced in different regions (39, 48) and if these correspond to regions more susceptible to strain injury (49). Therefore, the purpose of this study is to investigate the effects of a 6-week intervention matching volume and eccentric contraction time between the RDL and the NHE on regional BFlh architecture and hamstring morphology. We hypothesized that the RDL would induce greater BFlh FL increases (and concurrent reductions in PA) proximally compared to the NHE, while the NHE will have greater FL increases (and concurrent reductions in PA) changes distally compared to the RDL. Secondly, we hypothesized there would be no changes in regional ACSA across the hamstring muscles.

METHODS

Trial Design

The Health Sciences Institutional Review Board at the University of Wisconsin-Madison approved all procedures and participants provided written informed consent prior to enrollment (IRB ID: 2022-0763). The trial was pre-registered on ClinicalTrials.gov (Identifier: NCT05455346) and the full methodology was previously published (50). The study was a single-blinded, randomized intervention trial, where participants were randomly allocated to either the RDL or NHE intervention group using a permuted block randomization (1:1 intervention group

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allocation). Following baseline outcome measures and randomization, participants underwent two supervised training sessions per week for six weeks according to their group allocation (see *Intervention*). Following the 6-week intervention, all baseline outcome measures were repeated. Participants then underwent a 2-week detraining period followed by a final assessment of all study outcome measures (Figure 1). Since fascicle length changes are known to return to baseline values within 2 weeks after the cessation of NHE (16, 21), repeat imaging was performed between 24 and 72 hours after the final training session or the 2 week detraining date. The Consolidated Standards of Reporting Trials (CONSORT) guidelines are included in the Supplemental Digital Content (see Supplemental Table, Supplemental Digital Content 1).

Inclusion Criteria and Subject Recruitment

Participants were recruited and enrolled from the university student body and university club soccer teams from November 2022 to September 2023. All study procedures were completed in December 2023. Inclusion criteria included: aged 18 to 25 years, self-report of being physically active—as defined by the Physical Activity Guidelines for Americans (51), having at least six months of experience in resistance training, no history of hamstring strain injury within the last six months, no lifetime history of lower extremity surgery, no current lower limb musculoskeletal injury, and females not currently pregnant. Participants were permitted to continue with their usual exercise and physical activity habits but were instructed to avoid any additional hamstring-specific resistance training, sprint exposure, and initiating any new exercise activities.

All data were collected in a dedicated laboratory space. Ultrasound measures were collected in a private room, and all training sessions were performed in a dedicated training facility

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within the laboratory space and under the supervision of lab personnel (JV, QT, CS) with at least four years of exercise testing and prescription experience.

An *a priori* power calculation was performed using R statistical software and the "WebPower" package (wp.kanova function) (50, 52). Using an effect size derived from several relevant studies investigating the effects of NHE on the primary aim of BFlh fascicle length changes (Cohen's f = 0.8), $\alpha = 0.05$, and power = 0.80, a total sample size of 11 participants per group was deemed appropriate (16, 17, 53–55). A total of 16 participants per group (N = 32 in total) were recruited based on sample sizes required for secondary measures and a 10% attrition rate (50).

Intervention

The detailed description of the intervention was reported previously (50). Briefly, participants underwent a standardized warm-up protocol prior to each training session that included a 5-minute general warm-up on a stationary bike followed by dynamic exercises that targeted both upper and lower body. Following the general warm-up, participants then performed their allocated hamstring exercise at submaximal intensities of 50, 75, and 85-90% for 10, 5, and 5 repetitions at each intensity, respectively.

Participants were given the exact same total body resistance program for the entire six-week intervention with the only difference being the interventional hamstring-specific exercise (either RDL or NHE) (50). The only lower body exercise performed during the intervention was the hamstring-specific exercise based on group allocation. Other upper body exercises included

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incline dumbbell bench press, chest-supported incline dumbbell rows, prone shoulder Y-T-W raises, biceps curls to triceps press downs, and crunches. These exercises were prescribed to encourage intervention compliance, minimize the amount of outside resistance training, and control for potential confounding factors that might contribute to changes in strength (e.g., improved scapular control and back strength that could influence RDL 3RM strength). The program was structured such that the first 2 weeks were a familiarization period followed by 4 weeks of progressive training. During the familiarization period, participants progressively increased the number of hamstring exercise repetitions (from 3 sets of 3 repetitions in Week 1 to 3 sets of 4 repetitions in Week 2) and relative intensity (from rate of perceived exertion (RPE) of 5 in Week 1 to RPE of 7 in Week 2) to prepare for the 4 week progressive training. During the 4 week progressive training, the same number of repetitions (4 sets of 4 repetitions) were performed at the same relative intensity (RPE of 8-9). The full intervention program can be found in the Supplemental Digital Content (see Supplemental Table, Supplemental Digital Content 2).

The hamstring exercises were performed with the same number of repetitions completed between the RDL and NHE groups. The intensity was also matched based on participants' rate of perceived exertion (RPE) on a 0 to 10 scale, where relative intensity was maintained at an RPE of 7 during the familiarization period and an RPE of 8-9 for the 4-week training period. Wrist straps were worn during the RDL testing and training sessions to ensure the training loads were not limited by the participant's grip strength. Finally, the exercise tempo was also matched between groups and progressed such that the eccentric contraction duration was 4 seconds for the first week, 5 seconds for the second week, and 6 seconds for the final four weeks. A metronome (60 Hz) was used during the hamstring exercise to provide audible feedback for execution of the eccentric

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contraction duration throughout the full range of motion of the RDL or NHE. Each repetition for the RDL was performed at the prescribed tempo for the eccentric phase with participants then returning to the starting position (concentric phase).

Since progressive overload is an important factor in eliciting architectural changes, the relative training intensity was maintained at an RPE of 8-9 by adding resistance to the respective exercise (18, 56). External load was added to the barbell for the RDL group. If a participant was able to control the NHE within the last ~20° perpendicular to the ground, then the relative intensity during the NHE was maintained by having participants hold weight plates across their chest (18). Loads were incrementally increased by ~2.5 kg to maintain the RPE of 8-9.

Ultrasound B-mode Image Acquisition

Considering the user-dependent nature of ultrasound imaging (57), the test-retest reliability of the ultrasound measures were determined in a separate cohort of participants with the same inclusion criteria as those enrolled for the intervention (N = 10) . For both reliability and the main study procedures, participants were positioned prone on an exam table with their hips and knees in a neutral position and feet off the end of the exam table. Participants laid quietly at rest for 3 min prior to ultrasound image acquisition to normalize fluid shift within the muscle (58, 59). All ultrasound images were collected using a Logiq P9 ultrasound system (GE Healthcare, Waukesha, WI) and a linear array transducer (L3-12-RS, 47.1 mm aperture). One researcher (SKC) acquired all images across all study time points and was blinded to group allocation for all participants.

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Imaging was performed unilaterally on the dominant limb, which was determined by asking the individual which foot they would use to kick a soccer ball as hard as they could. Three separate longitudinal extended-field-of-view (EFOV) images of the entire BFlh muscle were collected from the most proximal to the most distal visualization of the muscle-tendon junctions.

A single researcher (CS) performed all architectural measures for the test-retest reliability, while another researcher (QT) performed all ACSA measures for the test-retest reliability. The test-retest reliability was overall excellent (ICC(3,1) > 0.90) for architectural (fascicle length (FL) ICC = 0.96, pennation angle (PA) ICC = 0.96, and muscle thickness (MT) ICC = 0.94) and morphological measures (ICC = 0.95). Reliability for FL, PA, and MT was good (0.75 < ICC < 0.90) to excellent across all regions of the BFlh (ICC(3,1): proximal [FL, PA, MT] = [0.94, 0.98, 0.77], mid-belly = [0.93, 0.96, 0.98], and distal = [0.88, 0.93, 0.94]). The FL coefficient of variation (CV) was 2.2%, 2.7%, and 4.8% for the proximal, mid-belly, and distal regions, respectively. The PA CV was 3.4%, 4.5%, and 6.2% for the proximal, mid-belly, and distal locations, respectively. The MT CV was 3.4%, 1.2%, and 3.4% for the proximal, mid-belly, and distal locations. The minimal detectable change (MDC) for regional FL was 0.8 cm, 0.8 cm, and 1.0 cm for proximal, mid-belly, and distal locations, respectively. The MDC for the regional PA was 1.3°, 1.7°, and 1.8° for the proximal, mid-belly, and distal locations, respectively. The MDC for the regional MT was 0.2 cm, 0.1 cm, and 0.2 cm, respectively.

Reliability for ACSA was excellent regardless of muscle (ICC(3,1) for BFlh = 0.95, ST = 0.98, SM = 0.98) or location (ICC(3,1): proximal = 0.94, mid-belly = 0.93, distal = 0.98). The ACSA CV was 6.4%, 2.2%, and 4.9% for the BFlh in the proximal, mid-belly, and distal regions,

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respectively. The ACSA CV was 8.2%, 8.4%, and 9.3% for the ST in the proximal, mid-belly, and distal regions, respectively. The ACSA CV was 6.8%, 6.0%, and 2.9% for the SM in the proximal, mid-belly, and distal regions, respectively. The MDC for BFlh was 3.1 cm^2, 1.3 cm^2, and 1.8 cm^2 for the proximal, mid-belly, and distal regions, respectively. The MDC for ST was 3.4 cm^2, 3.9 cm^2, and 1.8 cm^2 for the proximal, mid-belly, and distal regions, respectively. The MDC for SM was 1.0 cm^2, 2.3 cm^2, and 1.1 cm^2 for the proximal, mid-belly, and distal regions, respectively.

Primary and Secondary Outcome Measures: Image Analysis

All architectural and morphological measures were analyzed using ImageJ software and repeated for baseline, post-intervention, and detraining study time points. The pixel to centimeter conversion ratio for each image was calibrated prior to extracting primary and secondary outcome measures.

The primary outcome variable was FL assessed across the different regions along the BFlh. The FL was determined by tracing three representative fascicles within each of the proximal, mid-belly, and distal regions using the segmented line tool (Figure 2A). Preliminary analyses in the separate reliability cohort showed <1.9% difference in architectural measures when comparing the mean of three fascicles from the best quality longitudinal EFOV image compared to the mean of three fascicles from each of the three longitudinal EFOV images. This difference corresponded to 0.1 cm, 0.2°, and 0.04 cm measurement differences for FL, PA, and MT, respectively. Therefore, FL and PA were measured from three representative fascicles in each region and MT was measured at the proximal, mid-belly, and distal locations from the best quality EFOV image.

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The secondary outcome measures included regional PA, muscle thickness, and anatomical cross-sectional area (ACSA). Pennation angle was derived from the same 3 fascicles from which the FL was measured and was determined as the angle between the fascicle and the intramuscular aponeurosis (Figure 2B). Muscle thickness was calculated as the vertical length between the superficial and intramuscular aponeuroses. Muscle thickness measures were taken at 20, 50, and 80% of the image width and subsequently averaged to try and account for any tapering of the muscle, particularly in the distal region (Figure 2C). The same researcher (CS) performed all architectural measures across muscle locations and study time points. This rater had an intra-rater reliability (ICC(3,1)) of 0.926 (0.853, 0.964), 0.785 (0.593, 0.892), and 0.916 (0.833, 0.959), respectively, for FL, PA, and MT in a training set of images (180 images).

Regional ACSA was calculated for BFlh, ST, and SM muscles. Each hamstring muscle was manually segmented by identifying the fascial borders and circumscribing the muscle (Figure 3). The area feature in ImageJ was used to calculate the area. These procedures were repeated for the 3 EFOV transverse images acquired at each location along the thigh and averaged for each location along the muscle. The same rater (JV) performed all ACSA measures and had an intra-rater reliability of ICC(3,1) = 0.971 (0.956, 0.981) in a training set of transverse EFOV images.

Statistical Analyses

An intention-to-treat method was used for all analyses. Separate linear mixed effects models were performed for the primary outcome (FL) and each of the secondary outcomes (PA, MT, and ACSA). For architectural measures of the BFlh (FL, PA) and muscle thickness, full factorial models were implemented with time (baseline, post-intervention, detraining), muscle

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location (proximal, mid-belly, distal), and intervention group (RDL, NHE) modeled as fixed effects and participant modeled as a random effect. For ACSA, full factorial models were implemented with time, muscle location, intervention group, and muscle (BFlh, ST, and SM) modeled as fixed effects and participant modeled as a random effect. If an interaction was significant, only comparisons made within the same region across time or across groups are presented. If an interaction was not significant, the interaction was removed, and the analysis was repeated with main effects only. *Post hoc* pairwise comparisons with Tukey corrections were made for significant interactions and main effects. Data are presented as least square mean estimate differences [95% confidence interval]. All statistics were run in RStudio software and *a priori* significance set at $\alpha = 0.05$.

RESULTS

Demographic Information

A total of 32 participants were enrolled in the study (Figure 1). The demographic information for all study participants are shown in Table 1. The average compliance rate for the hamstring intervention was 11.5 out of 12 sessions (95%, range: 10-12) for the RDL group and 11.3 out of 12 sessions (94%, range: 9-12) for the NHE group.

Biceps Femoris Long Head Fascicle Length

There was a significant location by group interaction on BFlh FL ($p = 0.01$), but no other interactions were significant ($p > 0.59$). When inspecting pairwise comparisons between the same location, there were no significant differences between groups at the proximal ($p = 0.99$), mid-belly ($p = 0.89$), or distal ($p = 0.91$) locations. Therefore, the model was run with main effects only.

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Since no significant interactions were detected, main effects were tested with least squares means differences between time points, locations, and groups being reported. There was a significant effect of time on BFlh FL ($p < 0.001$) with *post hoc* pairwise comparisons showing a significant increase in FL from baseline to post-intervention (9.0 [8.7, 9.3] vs. 9.8 [9.5, 10.2] cm, mean difference = 0.80 [0.6, 1.0] cm, $p < 0.001$) and a significant decrease from post-intervention to detraining (9.8 [9.5, 10.1] vs. 9.1 [8.8, 9.4] cm; mean difference = -0.72 [-0.9, -0.5] cm, $p < 0.001$) (Figure 4). There were no differences in FL between baseline and detraining (mean difference = -0.07 [-0.3, 0.1], $p = 0.63$).

There was also a significant effect of location on BFlh FL ($p < 0.001$). P*ost hoc* pairwise comparisons showed longer FL proximally compared to mid-belly (11.2 [10.9, 11.5] vs. 9.1 [8.8, 9.4] cm, mean difference = 2.1 [1.9, 2.3] cm, $p < 0.001$) and distal regions (11.2 [10.9, 11.5] vs. 7.7 [7.4, 8.0] cm, mean difference = 3.5 [3.4, 3.7] cm, $p < 0.001$). Mid-belly FL was also longer than distal FL (mean difference = 1.4 [1.2, 1.6] cm, $p < 0.001$). There were no differences between groups in FL (NHE = 9.4 [9.0, 9.8] vs. RDL = 9.2 [8.8, 9.7] cm, mean difference = 0.17 [-0.42, 0.76] cm, $p = 0.55$).

Biceps Femoris Long Head Pennation Angle and Muscle Thickness

There were no significant interactions for PA ($p > 0.32$). There was a significant main effect of time ($p < 0.001$) with *post hoc* pairwise analysis showing a significant decrease in PA from baseline to post-intervention (11.5° [11.0, 12.0] vs. 10.3° [9.8, 10.8], mean difference = -1.2° [-1.5, -0.9], $p < 0.001$) and a significant increase in PA from post-intervention to detraining (10.3° [9.8, 10.8] vs. 11.2° [10.7, 11.7], mean difference = 1.0° [0.7, 1.3], $p < 0.001$) (Figure 5). There

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were no differences in PA between baseline and detraining (mean difference = 0.3° [-0.04, 0.57], p = 0.11). There was also a significant effect of location on PA ($p < 0.001$). P*ost hoc* pairwise comparisons showed smaller PA proximally compared to mid-belly (11.9° [11.4, 12.4] vs. 12.5° [12.0, 13.0], mean difference = -0.6° [-0.9, -0.3], $p < 0.001$) but larger PA proximally compared to the distal region (11.9° [11.4, 12.4] vs. 8.7° [8.2, 9.2], mean difference = 3.1° [2.8, 3.4], $p < 0.001$). Mid-belly PA was also larger than distal PA (mean difference = 3.7° [3.4, 4.0], $p < 0.001$). There were no differences between groups in PA (NHE = 11.1° [10.4, 11.8] vs. RDL = 10.9° [10.3, 11.6], mean difference = 0.16° [-0.81, 1.12], p = 0.74).

There were no significant interactions on MT ($p > 0.32$). There was a significant main effect of location ($p < 0.001$) with *post hoc* pairwise analysis showing the proximal region was thicker than the distal (2.2 [2.1, 2.3] vs. 2.0 [1.9, 2.1] cm, mean difference = 0.2 cm [0.17, 0.27], $p < 0.001$) but smaller than the mid-belly region (2.2 [2.1, 2.3] vs. 2.3 [2.2, 2.4] cm, mean difference = -0.1 [-0.15, -0.04] cm, $p < 0.001$) (Figure 5). The mid-belly was also thicker than the distal location (mean difference = 0.3 cm [0.26, 0.36] $p < 0.001$). There were no differences in MT across time (p = 0.29). There were no differences between groups in MT (NHE = 2.22 [2.1, 2.4] vs. RDL = 2.16 [2.0, 2.3], mean difference = -0.06 [-0.12, 0.25] cm, p = 0.48).

Hamstrings Anatomical Cross-Sectional Area

There was a significant muscle by location interaction ($p < 0.001$) for ACSA, but no other interactions were significant ($p > 0.07$). The least square means for each muscle and location ACSA is given in Figure 6. The proximal region of the BFlh muscle had a significantly smaller ACSA compared to the mid-belly (mean difference = -3.0 cm^2 [-3.8, -2.3], $p < 0.001$) but not the distal

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region (mean difference = -0.1 cm^2 [-0.84, 0.69], p = 1.00). The BFlh ACSA at mid-belly was significantly greater than the distal region (mean difference = 3.0 cm^2 [2.2, 3.7], $p < 0.001$). The proximal region of the ST muscle had a significantly larger ACSA compared to the mid-belly (mean difference = 2.5 cm^2 [1.8, 3.3], $p < 0.001$) and distal regions (mean difference = 4.5 cm^2 [3.8, 5.3], $p < 0.001$). The ST ACSA at mid-belly was significantly larger than the distal region (mean difference = 2.0 cm^2 [1.2, 2.8], $p < 0.001$). The proximal region of the SM had a significantly smaller ACSA compared to the mid-belly (mean difference = -6.3 cm^2 [-7.1, -5.6], $p < 0.001$) and distal regions (mean difference = -7.6 cm^2 [-8.4, -6.8], $p < 0.001$). The SM ACSA at mid-belly was significantly smaller compared to the distal region (mean difference = -1.3 cm^2 [-2.0, -0.5], $p < 0.001$). The complete set of pairwise comparisons are given in the Supplemental Digital Content (see Supplemental Table, Supplemental Digital Content 3).

Since no additional significant interactions were detected, main effects were tested with least squares means differences between time points, locations, muscles, and groups being reported. There was also a significant main effect of time where ACSA was significantly larger after the intervention (baseline to post intervention: 8.2 [7.6, 8.8] vs. 8.9 [8.3, 9.5] cm^2, mean difference = 0.78 cm^2 [0.45, 1.11], $p < 0.001$) but decreased between post-intervention and detraining (mean difference = -0.43 cm^2 [0.10, 0.76], p = 0.007) (Table 2). The ACSA remained greater at detraining compared to baseline (baseline to detraining: 8.2 [7.6, 8.8] vs. 8.5 [7.9, 9.1] cm^2, mean difference = 0.35 cm^2 [0.01, 0.68], p = 0.04). There were no differences between groups in ACSA (NHE = 8.4 [7.6, 9.2] vs. RDL = 8.7 [7.9, 9.5] cm^2, mean difference = 0.30 [-0.86, 1.5] cm^2, p = 0.60).

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DISCUSSION

This study investigated the effects of a 6-week eccentrically-biased RDL or NHE intervention on regional BFlh architecture and hamstring morphology. Both eccentrically-biased exercise interventions were effective at increasing FL, decreasing PA, and increasing ACSA without differences between the RDL and NHE for any architectural or morphology measure.

Many studies have investigated the effects of eccentric hamstring strength training on injury risk factors, with the NHE being widely investigated (15). The NHE exercise is effective in increasing BFlh FL, decreasing BFlh PA, and reducing HSI incidence (15–18). Considering the success of the NHE in reducing HSI injury rates and inducing beneficial architectural adaptations, it is clear why many studies use the NHE as the reference against which other hamstring exercises are measured. An alternative knee-dominant hamstring exercise, known as the razor hamstring curl, does not induce architectural adaptations of the BFlh muscle after 6-weeks of training like the NHE (18). When comparing the NHE to hip-dominant hamstring exercises, Bourne et al. (17) found the 45° hip extension resulted in similar BFlh FL increases to the NHE. Timmins et al. (34) compared the NHE against a flywheel RDL intervention and observed that BFlh FL increased while BFlh PA decreased similarly in both groups across a season-long (39 weeks) intervention. As a follow-up study, Timmins et al. (60) compared the traditional eccentrically-biased NHE to an isometric hold in a mid-repetition NHE position and observed no differences between groups in FL or PA after the pre- and in-season training. Similar to these studies, we did not observe differences between the RDL and NHE intervention groups in any architectural or morphological measure indicating our findings are in agreement with these previous investigations.

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Our study builds upon previous investigations by simultaneously investigating the effects of the NHE and RDL on regional BFlh architecture and morphology. Most investigations have only acquired architectural measures to the mid-belly of the BFlh which does not account for within-muscle architectural variability (17, 18, 34, 39, 61). Although we did observe a significant location by group interaction in FL, these differences were primarily dictated by the inherent regional differences in FL along the BFlh muscle (39), since there were no differences in any region when comparing the same locations across groups. Specifically, BFlh FL was longest in the proximal region, with the mid-belly region also having longer fascicles than the distal region. These findings complement previous investigations in both cadavers and human subjects, which showed that fascicle lengths were longer in the proximal region compared to mid-belly or distal regions (39, 41). We also observed that BFlh PA was largest in the mid-belly region, which is in agreement with previous investigations, though some discrepancies within the literature exist (39, 41, 61). One study found that PA was largest proximally (39), whereas data from other investigations showed BFlh had a smaller PA proximally compared to mid-belly (41, 61). Together, these data suggest that both the RDL and NHE increased FL and decreased PA to a similar extent across all three locations following the intervention.

Contrary to our hypothesis, FL and PA adaptations after the intervention were similar between groups along the BFlh muscle. Simulation and cadaveric studies demonstrate that the BFlh muscle undergoes greater muscle-tendon length changes with altered hip joint angles compared to alterations in knee joint angle (62). Though different exercises do not appear to induce different levels of adaptation at the mid-belly, it seemed plausible that differences may exist regionally in the BFlh after hip- or knee-dominant eccentrically-biased training. Our data suggest

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that 6-weeks of training with either the hip-dominant RDL or the knee-dominant NHE elicit similar architectural changes along the muscle. It is unclear if longer intervention durations of the RDL and NHE may elicit different regional changes in architecture, though FL and PA changes at the mid-belly appear similar between the NHE and other hamstring exercises over longer training durations (17, 34, 60).

We also investigated the changes in both regional muscle thickness and ACSA of the BFlh. Neither intervention increased BFlh muscle thickness over the 6-week period, but there were differences in muscle thickness between regions. The mid-belly region was the thickest part of the BFlh muscle with both the proximal and distal regions smaller than the mid-belly. This finding is consistent with previous ultrasound and cadaveric dissection studies (41). Many studies with varying training intervention duration (range: 6 to 39 weeks) have shown that muscle thickness does not change following NHE training (16, 18, 34, 60). It should be noted that across these studies (and the current investigation) muscle thickness was measured from the superficial to intramuscular aponeuroses. It is possible that muscle thickness of the superficial compartment of the BFlh may not reflect the totality of morphological change that accompanies resistance training, and that ACSA should instead be used to assess morphology changes after eccentrically-biased training.

The combined ACSA across all hamstring muscles increased after intervention and reduced after detraining, but still remained elevated compared to baseline. Previous investigations have characterized changes in ACSA after training and a period of detraining. In a small study of 4 male subjects, the authors found that ACSA of the quadriceps muscles increased by ~8.5% after 60 days

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of training but decreased at a nearly identical rate following detraining (63). Other studies in both young and middle-aged participants found ACSA increased following resistance training and returned to baseline values after a detraining period that matched the training period (64, 65). *Post hoc* analysis in the current investigation revealed that percentage change in ACSA after the intervention was ~10% across all hamstring muscles and decreased by ~50% after detraining (~5% increase relative to baseline). These results appear consistent with the mirrored time course of ACSA increases following training and subsequent decreases after a period of detraining.

Despite the significant increases in ACSA after 6 weeks of training, we did not observe differences in ACSA changes between groups. This is somewhat consistent with recent studies using magnetic resonance imaging (MRI) to assess ACSA changes after 10 weeks of NHE or the hip extension exercise (17, 66). The authors observed differences in both absolute and percentage change in cross-sectional area of BFlh and ST compared to controls in the mid-belly to distal regions (~30-70%). However, differences were not observed in maximal ACSA between training groups (17). In another investigation, lengthened state eccentric training (LEST) and hip extension exercise training were compared to the NHE. It was observed that both LEST and hip extension exercises promoted greater increases in overall muscle volume in the BFlh compared to the NHE (17, 67). It is likely that the null between-group findings in ACSA changes in the current study relative to these other studies could be due to the shorter duration of the intervention in our current study (6 vs. 10 to 12 weeks of training), unilateral training in both the LEST and hip extension exercises, measurement differences when comparing between ACSA and volume, and acquiring ACSA measures from ultrasound compared to MRI. Furthermore, the percentage change in ACSA after the intervention in the current study (~10% across all hamstring muscles) is lower compared

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to ACSA measures observed after 10 weeks of training (66), suggesting the longer training durations might elicit additional adaptations than observed herein.

There are some limitations of the study that should be acknowledged. The RDL exercise is not a fully eccentric exercise like the NHE. Therefore, the eccentrically-biased RDL likely underloads the hamstring muscles compared to the NHE, as participants were able to perform the concentric phase of the RDL to return to the starting position. Though it is possible to perform a purely eccentric RDL with supramaximal loads, thus eliminating the concentric return to the start of the movement, we chose to match the eccentric time-under-tension, volume, and relative intensity to more closely mimic how the RDL might typically be performed in practice. It is possible that a supramaximal eccentric RDL intervention might elicit greater adaptations than observed in the current study, but this is purely speculative and may be difficult to implement in practice. The 6-week duration of our study was shorter than previous studies that have observed larger changes in BFlh architecture and hamstring ACSA (17, 34, 60, 66, 67). It is unclear if a longer study duration would elicit greater changes in FL, PA, and ACSA than observed here or if regional differences in architecture and ACSA would be more pronounced.

CONCLUSIONS

In conclusion, the current study reports the effects of a 6-week eccentrically-biased RDL and NHE intervention on regional BFlh architecture and hamstring morphology. Increases in BFlh FL, decreases in BFlh PA, and increases in ACSA across all hamstring muscles following 6 weeks of training and 2 weeks of detraining are similar between these exercises. Differences in regional BFlh architecture were attributed to the within-muscle variation and not different training

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exercises. These findings provide practitioners with a choice in exercise selection between the RDL and NHE for eliciting adaptations thought to be beneficial for hamstring injury prevention measures.

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Acknowledgements

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FIGURE LEGENDS

Figure 1. Flowchart for participant inclusion.

Figure 2. Representative longitudinal extended-field-of-view image of the biceps femoris long head muscle. A) Muscle fascicle lengths (yellow) were determined from three fascicles in each of the proximal, mid-belly, and distal regions. B) Pennation angles (white) were determined from the insertion of the same three fascicles into the intramuscular aponeurosis. C) Muscle thickness (yellow vertical lines) was average across the proximal, mid-belly, and distal regions.

Figure 3. Representative images of segmentation of anatomical cross-sectional area at the A) proximal (33% thigh length), B) mid-belly, and C) distal (67% thigh length) regions.

Figure 4. Primary Outcome Measures of Biceps Femoris Long Head Fascicle Length Location along the Muscle and by Intervention Group. Squares represent the least square mean estimates with 95% confidence intervals. Fascicle length increased after intervention and then returned to baseline values following detraining with no differences between groups.

Figure 5. Secondary Outcome Measures of Biceps Femoris Long Head b) Pennation Angle, and c) Muscle thickness by Location along the Muscle and by Intervention Group. Squares represent the least square mean estimates with 95% confidence intervals. Pennation angle decreased after intervention followed by a return to baseline after detraining with no differences between groups. There were no changes in muscle thickness following intervention or between groups.

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Figure 6. Anatomical Cross-Sectional Area by Muscle and Location. Data are presented as least square mean [95% confidence interval]. * = notes within-muscle difference in ACSA ($p < 0.05$), † = notes differences from BFlh †, § = notes differences between ST and SM

SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Supplementary Material S1 - CONSORT Checklist.docx

SDC 2: Supplementary Material S2 - Intervention Program.docx

SDC 3: Supplementary Material S3 - ACSA LME Pairwise Comparisons.docx

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Figure 1

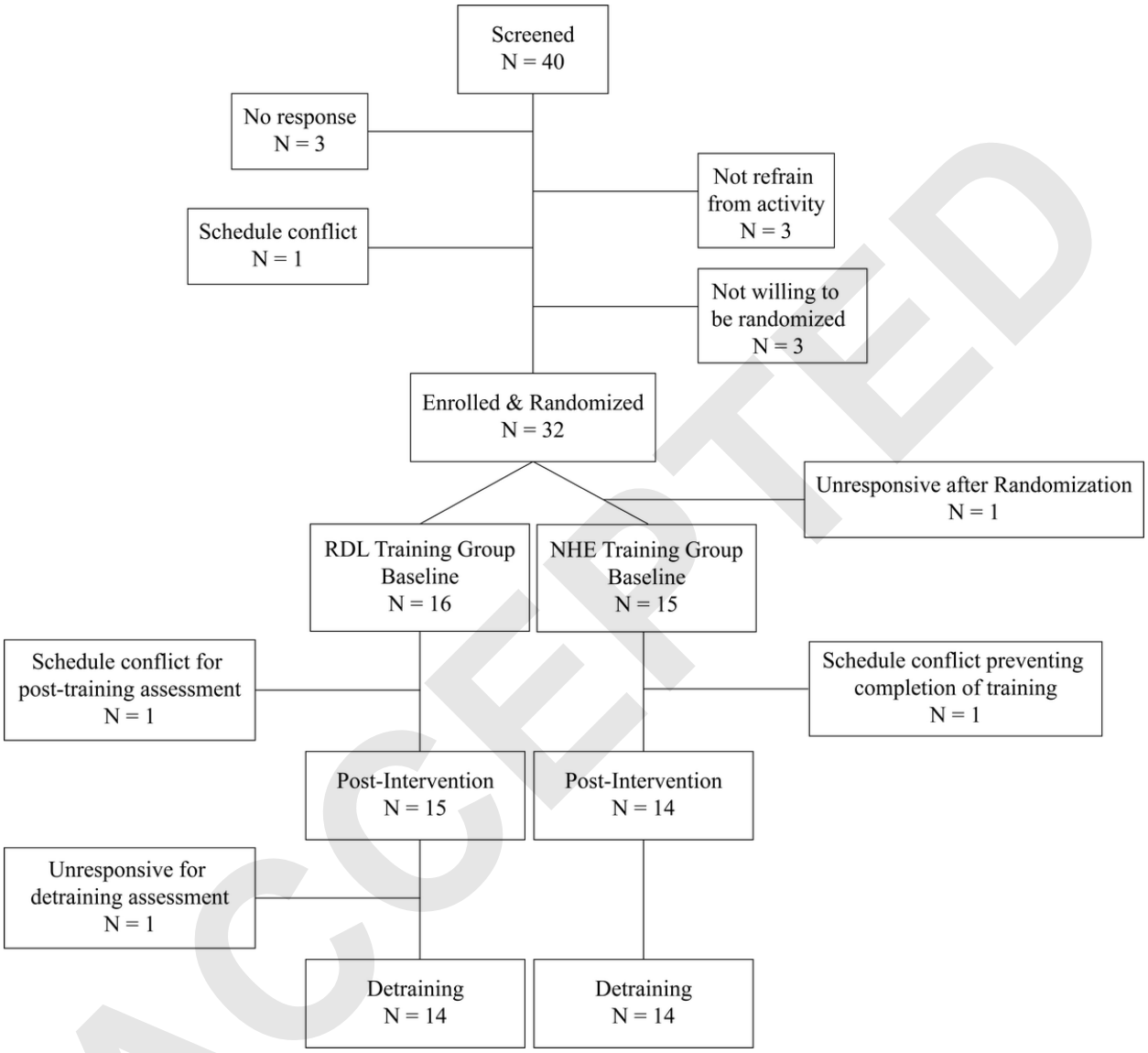

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Figure 2

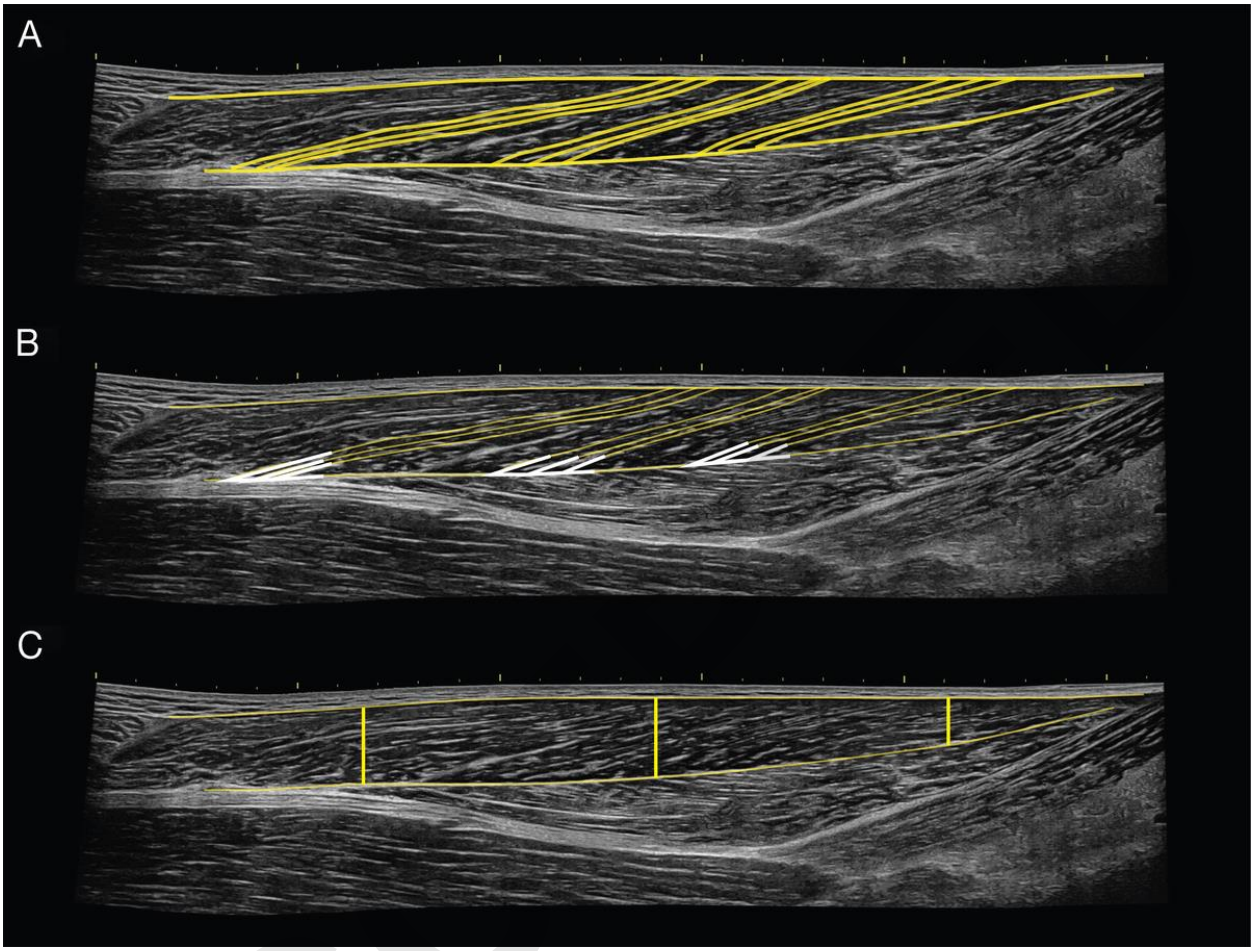

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Figure 3

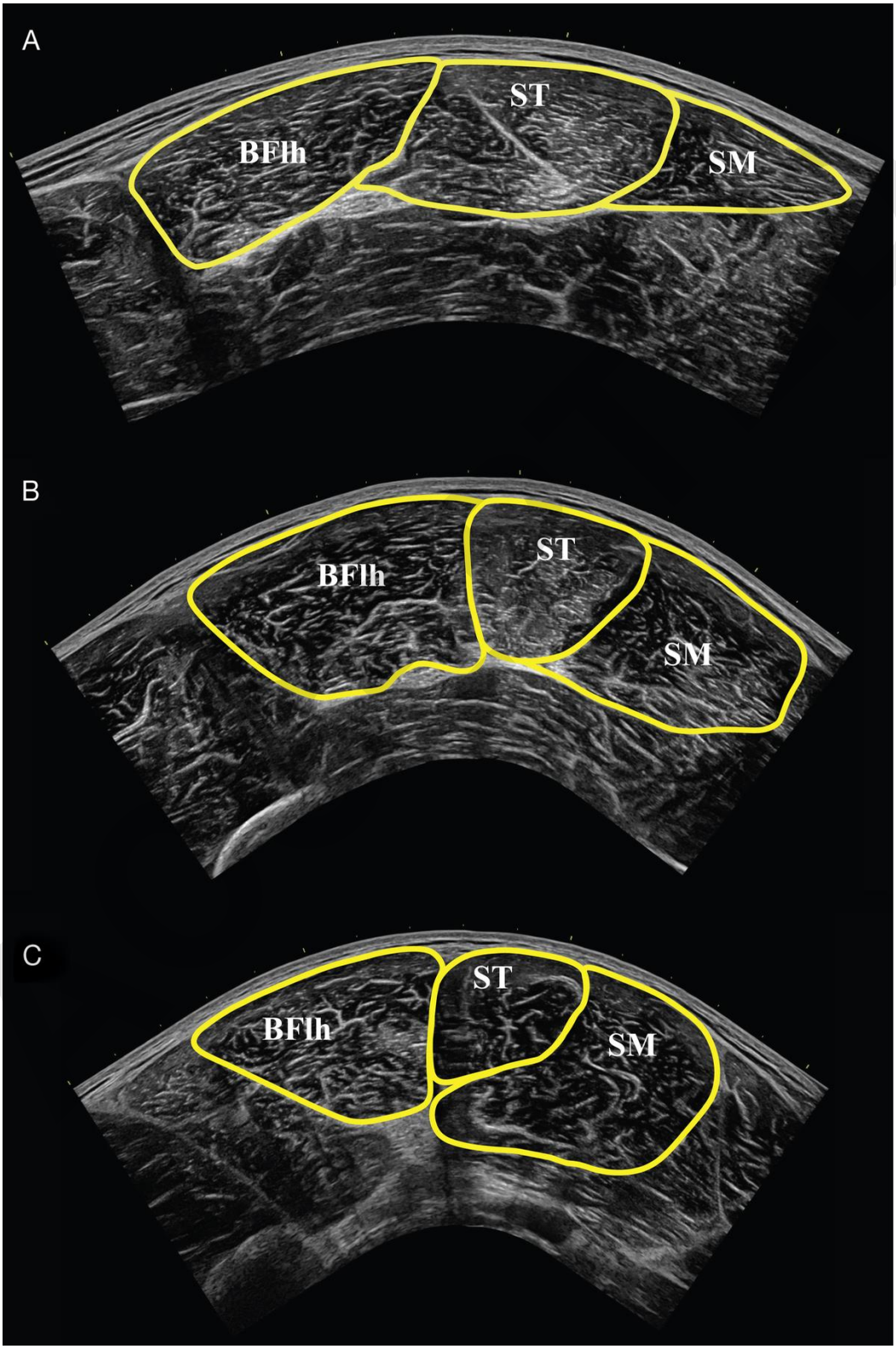

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Figure 4

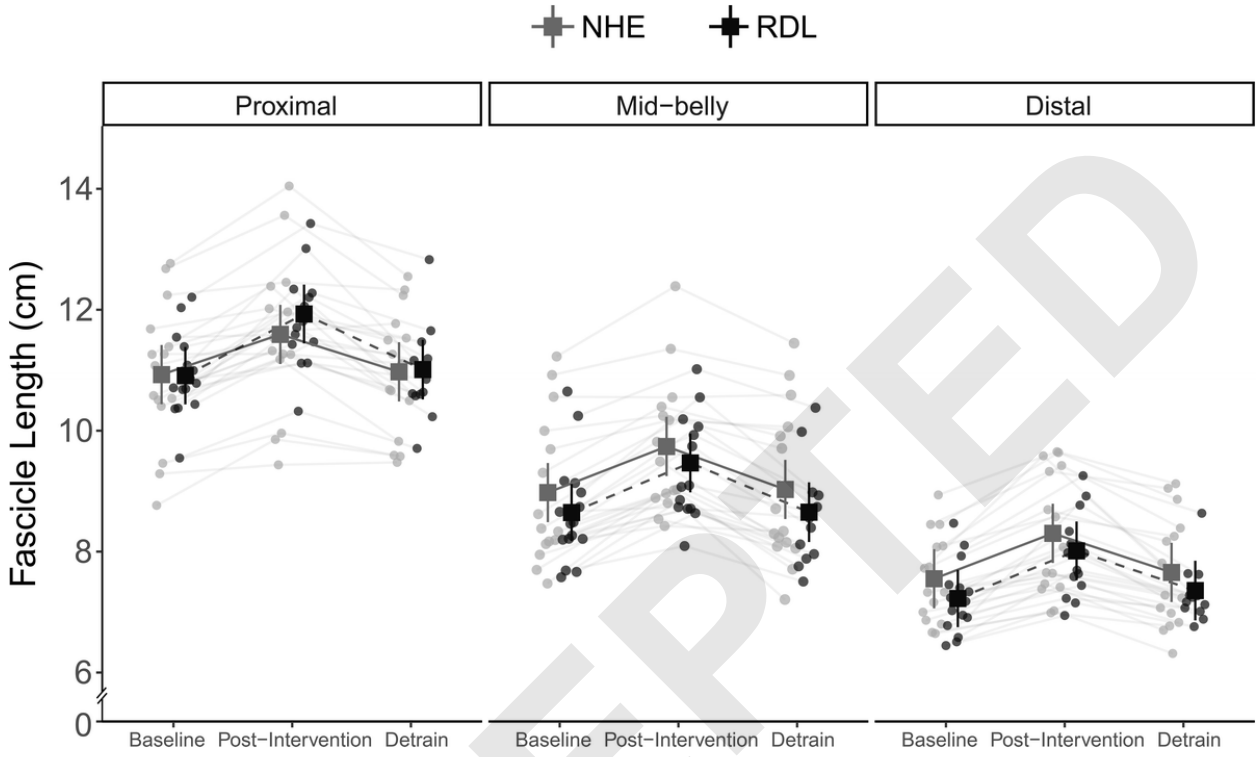

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Figure 5

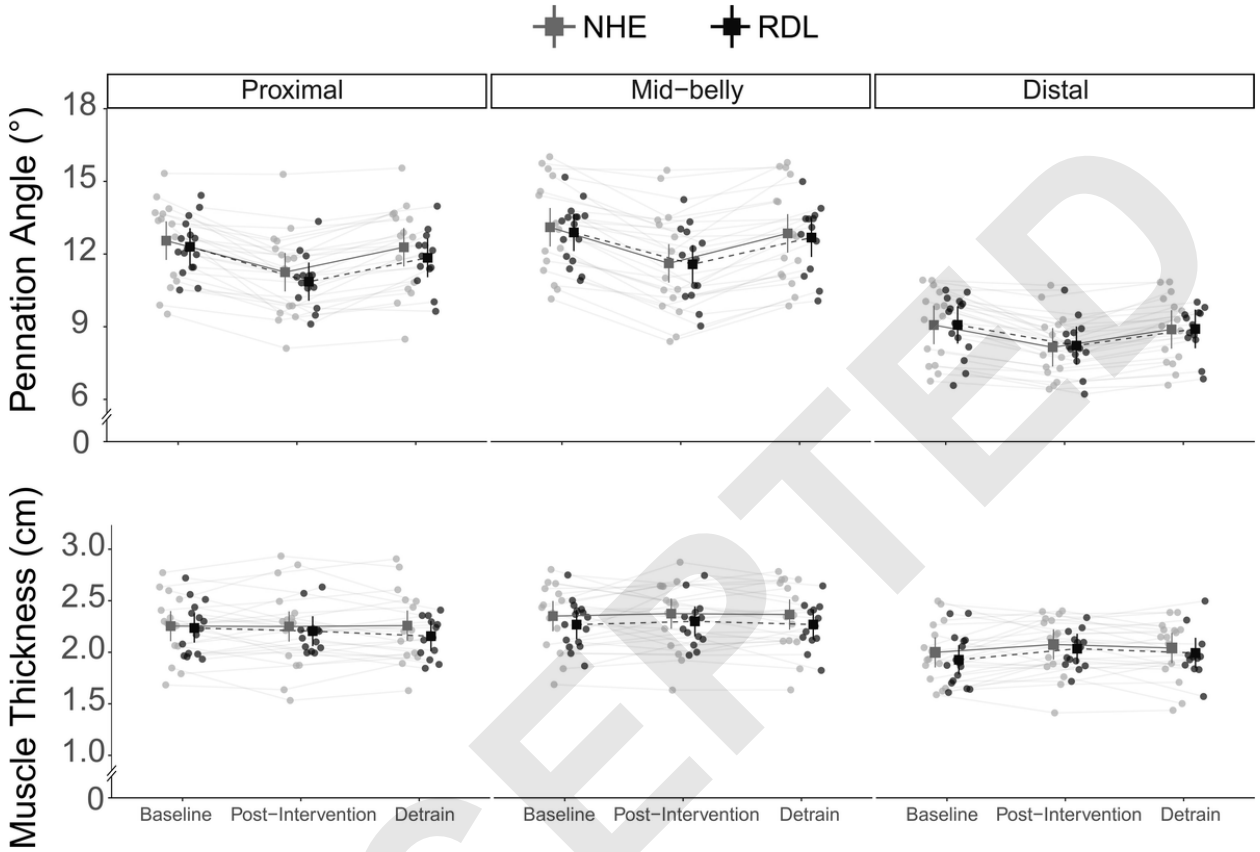

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Figure 6

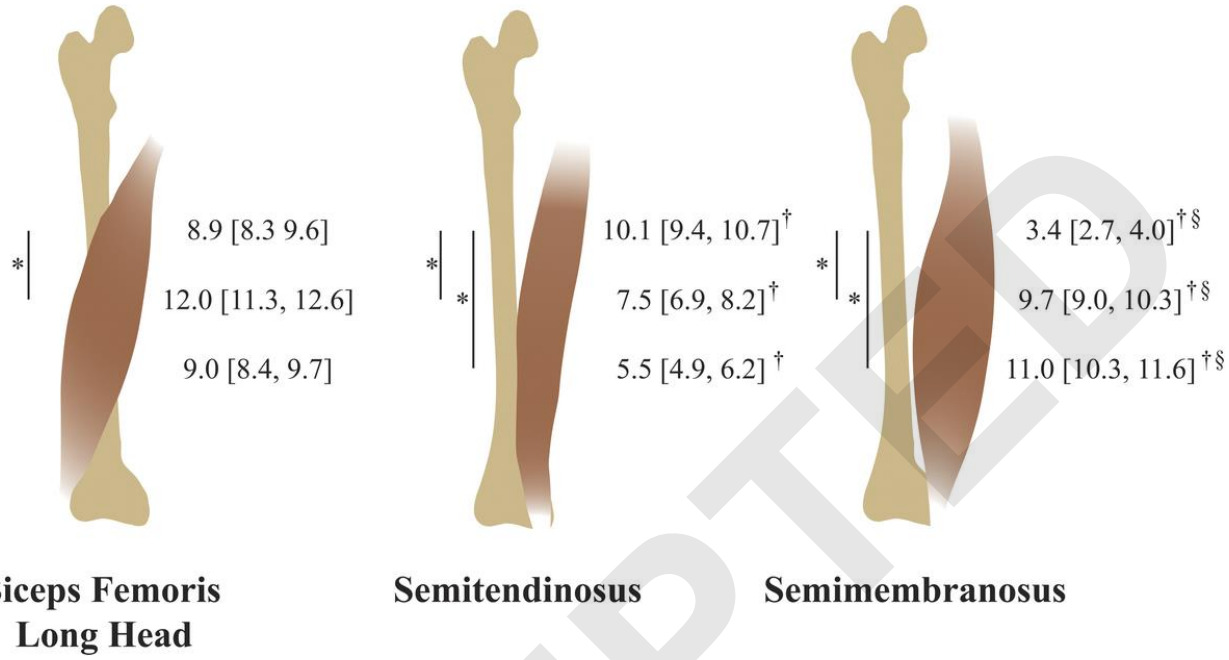

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Table 1. Participant demographics by intervention group. Data presented as counts or mean (standard deviation).

Participant Demographics	Intervention Group	
	NHE	RDL
Sex (F | M)	8 | 8	7 | 9
Age (years)	20.6 (0.9)	21.0 (0.8)
Height (cm)	171.4 (10.3)	173.6 (10.4)
Weight (kg)	69.5 (10.0)	70.1 (9.9)
Body mass index (kg/m^2)	23.6 (2.4)	23.2 (2.1)

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Table 2. Anatomical Cross-Sectional Area (ACSA) by Study Timepoint. Data are presented as pooled ACSA across all the hamstring muscles and as least square mean (cm^2) [95% confidence interval] and percent change (%).

Baseline	Post-intervention	Detraining	Baseline to Post-Intervention Change	Post-Intervention to Detraining Change	Baseline to Detraining Change
8.16 [7.57, 8.76]	8.94 [8.34, 9.54]	8.51 [7.91, 9.12]	9.9%	-4.5%	4.9%

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CONSORT 2010 checklist of information to include when reporting a randomised trial*

Section/Topic	Item No	Checklist item	Reported on page No
Title and abstract			
	1a	Identification as a randomised trial in the title	Title Page
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	1
Introduction			
Background and objectives	2a	Scientific background and explanation of rationale	3-5
	2b	Specific objectives or hypotheses	5
Methods			
Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	5
	3b	Important changes to methods after trial commencement (such as eligibility criteria), with reasons	NA
Participants	4a	Eligibility criteria for participants	6
	4b	Settings and locations where the data were collected	6
Interventions	5	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered	7-8
Outcomes	6a	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed	9-10
	6b	Any changes to trial outcomes after the trial commenced, with reasons	NA
Sample size	7a	How sample size was determined	6
	7b	When applicable, explanation of any interim analyses and stopping guidelines	NA
Randomisation:			
	8a	Method used to generate the random allocation sequence	5

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Sequence generation	8b	Type of randomisation; details of any restriction (such as blocking and block size)	5
Allocation concealment mechanism	9	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned	5
Implementation	10	Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions	5
Blinding	11a	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how	5,8
	11b	If relevant, description of the similarity of interventions	NA
Statistical methods	12a	Statistical methods used to compare groups for primary and secondary outcomes	10
	12b	Methods for additional analyses, such as subgroup analyses and adjusted analyses	10
Results			
Participant flow (a diagram is strongly recommended)	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analysed for the primary outcome	Figure 1, Table 1, pg. 11
	13b	For each group, losses and exclusions after randomisation, together with reasons	Figure 1
Recruitment	14a	Dates defining the periods of recruitment and follow-up	6
	14b	Why the trial ended or was stopped	NA
Baseline data	15	A table showing baseline demographic and clinical characteristics for each group	Table 1
Numbers analysed	16	For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups	11-16
Outcomes and estimation	17a	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval)	Figure 1
	17b	For binary outcomes, presentation of both absolute and relative effect sizes is recommended	NA

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Ancillary analyses	18	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory	NA
Harms	19	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms)	NA
Discussion			
Limitations	20	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses	18
Generalisability	21	Generalisability (external validity, applicability) of the trial findings	18
Interpretation	22	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	14-18
Other information			
Registration	23	Registration number and name of trial registry	5
Protocol	24	Where the full trial protocol can be accessed, if available	5
Funding	25	Sources of funding and other support (such as supply of drugs), role of funders	Disclosures

Citation: Schulz KF, Altman DG, Moher D, for the CONSORT Group. CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials. BMC Medicine. 2010;8:18.
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*We strongly recommend reading this statement in conjunction with the CONSORT 2010 Explanation and Elaboration for important clarifications on all the items. If relevant, we also recommend reading CONSORT extensions for cluster randomised trials, non-inferiority and equivalence trials, non-pharmacological treatments, herbal interventions, and pragmatic trials. Additional extensions are forthcoming: for those and for up-to-date references relevant to this checklist, see www.consort-statement.org.

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Supplemental Digital Content 1. Intervention Program.

The intervention program was structured such that participants underwent a 2-week familiarization period followed by 4 weeks of progressive loading. Training sessions were performed twice per week in a dedication training space and under the supervision of study personnel. The hamstrings exercise was performed per the intervention group allocation. Relative intensity determined as rate of perceived exertion (RPE) for the hamstrings exercise was increased from 5 to 7 (out of 10) for the first 2 weeks (familiarization) and then maintained ≥ 8 for all remaining training sessions. Training loads were recorded for each training session and exercise.

Week 1 (Familiarization)

Exercise	Sets x Reps	Target Intensity (0 to 10 RPE)	Tempo (ECC:ISO:CON)	Load		Inter-set Rest
				Day 1	Day 2	
Hamstrings	3 x 3	5	4:0:1			90 sec
Incline DB Chest Press	2 x 10	7	2:0:1			60 sec
Chest Supported Incline DB Rows	2 x 10	7	2:0:1			60 sec
Incline Bench Shoulder Y-T-W	2 x 10	7	2:0:1			60 sec
Bicep Curls into Triceps Press-down	2 x 15	7	2:0:1			60 sec
Half-Up Half-Down Crunches	2 x 10 each	BW	1:1:1	BW		60 sec

Week 2 (Familiarization)

Exercise	Sets/Reps	Target Intensity (0 to 10 RPE)	Tempo (ECC:ISO:CON)	Load		Inter-set Rest
				Day 1	Day 2	
Hamstrings	3 x 4	7	5:0:1			90 sec
Incline DB Chest Press	2 x 10	7	2:0:1			60 sec
Chest Supported Incline DB Rows	2 x 10	7	2:0:1			60 sec
Incline Bench Shoulder Y-T-W	2 x 10	7	2:0:1			60 sec
Bicep Curls into Triceps Press-down	2 x 15	7	2:0:1			60 sec

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Half-Up Half-Down Crunches	2 x 10 each	BW	1:1:1	BW	60 sec

Weeks 3-6 (Progressive Training)

Exercise	Sets/Reps	Target Intensity (0 to 10 RPE)	Tempo (ECC:ISO:CON)	Load		Inter-set Rest
				Day 1	Day 2	
Hamstrings	4 x 4 @ 6 sec	≥ 8	6:0:1			120 sec
Incline DB Chest Press	3 x 10	7	2:0:1			60 sec
Chest Supported Incline DB Rows	3 x 10	7	2:0:1			60 sec
Incline Bench Shoulder Y-T-W	3 x 10	7	2:0:1			60 sec
Bicep Curls into Triceps Press-down	3 x 12	7	2:0:1			60 sec
Half-Up Half-Down Crunches	3 x 10 each	BW	1:1:1	BW		60 sec

Abbreviations: BW = bodyweight, CON = concentric phase, DB = dumbbell, ECC = concentric phase, ISO = isometric phase, RPE = rate of perceived exertion

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Supplemental Digital Content 2. Anatomical Cross-Sectional Area Least Square Means by Muscle and Location.

Pairwise Contrast	Least Square Mean Difference	Lower Confidence Limit	Upper Confidence Limit	p-value
BFlh PROXIMAL - ST PROXIMAL	-1.133	-1.882	-0.385	0.0001
BFlh PROXIMAL - SM PROXIMAL	5.570	4.821	6.318	<0.0001
BFlh PROXIMAL - (BFlh MID-BELLY)	-3.035	-3.783	-2.287	<0.0001
BFlh PROXIMAL - (ST MID-BELLY)	1.397	0.649	2.145	0.0000
BFlh PROXIMAL - (SM MID-BELLY)	-0.736	-1.485	0.012	0.0578
BFlh PROXIMAL - BFlh DISTAL	-0.076	-0.827	0.675	1.0000
BFlh PROXIMAL - ST DISTAL	3.514	2.763	4.264	<0.0001
BFlh PROXIMAL - SM DISTAL	-2.020	-2.770	-1.269	<0.0001
ST PROXIMAL - SM PROXIMAL	6.703	5.955	7.451	<0.0001
ST PROXIMAL - (BFlh MID-BELLY)	-1.902	-2.650	-1.153	<0.0001
ST PROXIMAL - (ST MID-BELLY)	2.530	1.782	3.279	<0.0001
ST PROXIMAL - (SM MID-BELLY)	0.397	-0.351	1.145	0.7764
ST PROXIMAL - BFlh DISTAL	1.057	0.307	1.808	0.0005
ST PROXIMAL - ST DISTAL	4.647	3.896	5.397	<0.0001
ST PROXIMAL - SM DISTAL	-0.886	-1.637	-0.136	0.0078
SM PROXIMAL - (BFlh MID-BELLY)	-8.605	-9.353	-7.856	<0.0001
SM PROXIMAL - (ST MID-BELLY)	-4.173	-4.921	-3.424	<0.0001
SM PROXIMAL - (SM MID-BELLY)	-6.306	-7.054	-5.558	<0.0001
SM PROXIMAL - BFlh DISTAL	-5.646	-6.396	-4.895	<0.0001
SM PROXIMAL - ST DISTAL	-2.056	-2.807	-1.306	<0.0001
SM PROXIMAL - SM DISTAL	-7.589	-8.340	-6.839	<0.0001
(BFlh MID-BELLY) - (ST MID-BELLY)	4.432	3.684	5.181	<0.0001
(BFlh MID-BELLY) - (SM MID-BELLY)	2.299	1.550	3.047	<0.0001
(BFlh MID-BELLY) - BFlh DISTAL	2.959	2.209	3.710	<0.0001
(BFlh MID-BELLY) - ST DISTAL	6.549	5.798	7.299	<0.0001
(BFlh MID-BELLY) - SM DISTAL	1.016	0.265	1.766	0.0010
(ST MID-BELLY) - (SM MID-BELLY)	-2.133	-2.882	-1.385	<0.0001

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(ST MID-BELLY) - BFlh DISTAL	-1.473	-2.224	-0.722	0.0000
(ST MID-BELLY) - ST DISTAL	2.116	1.366	2.867	<0.0001
(ST MID-BELLY) - SM DISTAL	-3.417	-4.167	-2.666	<0.0001
(SM MID-BELLY) - BFlh DISTAL	0.660	-0.090	1.411	0.1367
(SM MID-BELLY) - ST DISTAL	4.250	3.499	5.001	<0.0001
(SM MID-BELLY) - SM DISTAL	-1.283	-2.034	-0.533	0.0000
BFlh DISTAL - ST DISTAL	3.590	2.837	4.342	<0.0001
BFlh DISTAL - SM DISTAL	-1.944	-2.696	-1.191	<0.0001
ST DISTAL - SM DISTAL	-5.533	-6.286	-4.781	<0.0001

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